MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION

October 2026 - Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMBG300 - AGRICULTURAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

(Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap))

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A- Short Notes Questions. Answer any 10 questions. Each question carries 2 marks

1. **Give four examples of soil algae. [U] / [CO1]**

- Chlorella , Anabaena , Chlamydomonas , Navicula , Oedogonium, Cylindrospermum, Nostoc, Tolypothrix. Any four.

2. **How does soil pH affect bacterial and fungal growth? [K] / [CO1]**

   Effects on Bacterial Growth

   • Preference for Neutral to Alkaline Soils: Most soil bacteria are neutrophiles that grow best in a neutral to slightly alkaline pH range.

   Effects on Fungal Growth

   • Preference for Acidic Soils: Fungi generally have wider optimal growth ranges and tolerate or thrive in lower (more acidic) pH environments compared to bacteria.

3. **Define predation. [K] / [CO1]**

   It is a biological interaction where one organism (the predator) hunts, kills, and consumes another organism (the prey) for food and energy.

4. **Define synergism with a suitable example. [K] / [CO1]**

   Association in which both populations benefit; a non-obligatory relationship. Any one example.

5. **Differentiate between phosphorus mineralisation and immobilisation. [K] / [CO2]**

Phosphorus mineralisation is the microbial conversion of organic phosphorus into available inorganic phosphate. In contrast, phosphorus immobilisation is the reverse process, in which soil microbes absorb inorganic phosphate and convert it into organic forms.

6. **What are epiphytic and endophytic microorganisms? [K] / [CO2]**

   Plant-associated microbes that differ primarily by where they live relative to the host plant.

   Epiphytic Microorganisms (Epiphytes)

   They live on the external surface of plants (such as leaves, stems, flowers, and fruits)

   Endophytic Microorganisms (Endophytes)

   They live inside the internal tissues of plants (such as within cells, intercellular spaces, or the vascular system) without causing visible disease.

7. **Distinguish between microbial retting and chemical retting. [An] / [CO2]**

   Microbial retting and chemical retting are two distinct methods used to extract plant fibres (such as jute, flax, and hemp) by breaking down the pectic substances that bind the fibres to the woody stem.

   Biological degradation using bacteria or fungi (anaerobic or aerobic) that secrete pectinolytic enzymes. Direct chemical breakdown using agents like sodium hydroxide.

8. **What is meant by root exudate? [K] / [CO2]**

   Root exudates are organic and inorganic fluids secreted or released by living plant roots into the surrounding soil, known as the rhizosphere. Any one component.

9. **How is sugarcane mosaic disease transmitted? [U] / [CO3]**

   • Aphid Vectors

   • Infected Planting Material

   • Mechanical Transmission. Any two

10. **Why does soft rot spread rapidly under warm and humid conditions? [U] / [CO3]**

    These environments speed up bacterial growth, increase plant moisture, and trigger the breakdown of cell walls.

11. **Summarise how applying lime to the soil helps control clubroot disease. [U] / [CO3]**

    Clubroot disease (caused by the pathogen Plasmodiophora brassicae) is controlled through two main mechanisms: raising soil pH and enriching the soil with calcium.

12. **State the main types of biopesticides. [K] / [CO3]**

    Bacterial, fungal and viral biopesticides.

[2x10 = 20]

Part B - Short Essay Questions. Answer any 4 questions. Each question carries 5 marks

13. Explain in detail Trichoderma as a biocontrol agent. [A] / [CO3]

- Biocontrol Definition (1 mark). Explain the role of Trichoderma in antagonism [1 mark], biocontrol [1 mark], induced systemic resistance [1mark], stimulation of plant growth and biodegradation. (1 mark)

14. Illustrate how Rhizobium inoculation can be used to improve the growth of a leguminous crop. [A] / [CO3]

Biofertilizer definition (1 mark); production (3 marks). Importance (1mark) • Supplies Usable Nitrogen • Promotes Growth Hormones • Enhances Nutrient Uptake: • Increases Yield and Soil Fertility:

15. Explain mutualism with suitable examples. [U] / [CO1]

It is a type of ecological interaction between two different species where both species benefit from the relationship. It is an obligatory and specific association. (1 mark). Lichen, mycorrhizae, symbiotic nitrogen fixation. Explain any one (4 marks)

16. Distinguish between ectomycorrhiza and endomycorrhiza. [U] / [CO1]

Key Differences

Feature	**Ectomycorrhiza**	**Endomycorrhiza (Arbuscular)**
Colonization	**Extracellular:** Fungi stay *outside* the root cells.	**Intracellular:** Fungi *penetrate* inside the root cortical cells.
Outer Sheath	Forms a dense outer fungal covering called a **mantle**.	Does **not** form a mantle.
Internal Network	Forms an intercellular network between cells called the **Hartig net**.	Forms tree-like **arbuscules** and storage **vesicles** inside the cells.
Fungal Partners	Mostly **Basidiomycota** and **Ascomycota**.	Mostly **Glomeromycota**.
Plant Hosts	Found in about 3%–10% of plants, mainly woody trees like pines, oaks, and spruces (conifers and hardwoods).	Found in about 80%–85% of plants, including most crops, grasses, vegetables, and flowers.

Root Alteration	Often makes the roots look swollen, stunted, or coral-like.	Does **not** change the outer shape of the roots.
Lab Culture	Can generally be grown (cultured) in a lab.	Cannot be cultured independently without a living host plant (obligate symbionts).

Any three points.

17. Describe the role played by microorganisms in the process of developing tobacco leaf quality during tobacco curing. [K] / [CO2]

Definition - tobacco curing (1 mark). Macromolecule Degradation. Reduction of Toxic and Irritating Compounds. Aroma and Flavour Precursor Generation. Explain each. (4 marks)

18. Explain the role of respiration in the carbon cycle. [A] / [CO2]

Returning Carbon to the Atmosphere: During aerobic respiration, living organisms break down glucose and other organic carbon compounds with oxygen to release energy (ATP). This process releases carbon dioxide back into the air or water. Explain the carbon cycle (5 marks)

[5x4 = 20]

Part C - Essay Question. Answer any 1 question. Each question carries 10 marks

19. Explain in detail the nitrogen cycle with the help of a diagram. [A] / [CO2]

Explain proteolysis(2 marks),ammonification(1 marks),nitrification(2 marks),denitrification(2 marks) and nitrogen fixation(2 marks) - Explain the role of microorganisms. Diagram(1 mark)

20. Explain how structural and biochemical defence mechanisms work together to protect plants from microbial infection. [A] / [CO3]

Structural and biochemical defence mechanisms –Definition and introduction (2 marks). Structural mechanisms (4 marks). Biochemical defence mechanisms (4 marks)

[10x1 = 10